

CS214: Computer Organization

Quiz 1

Time: 20 Minutes

Maximum Marks: 15

Instructions: For MCQs, write the selected option (A/B/C/D) in the MCQ answer box. For subjective questions, write all answers inside the designated answer boxes. Answers written outside the designated spaces will not be evaluated.

Question	Q1	Q2	Q3	Q4	Q5
Answer					

Q1.

[1 Mark]

Three positive-edge-triggered T flip-flops have outputs Q_2, Q_1, Q_0 . All three flip-flops are driven by the same clock, with

$$T_0 = 1, \quad T_1 = Q_0, \quad T_2 = Q_0 Q_1.$$

Initially, $Q_2 Q_1 Q_0 = 101$. The state of the circuit immediately after three active clock edges is

(A) 000 (B) 001 (C) 111 (D) 100

Q2.

[1 Mark]

A 4-bit register $Q_3 Q_2 Q_1 Q_0$ performs the following operation on each positive clock edge:

$S_1 S_0$	Operation	$S_1 S_0$	Operation
00	Hold	01	Shift right
10	Shift left	11	Parallel load

During a right shift, 0 enters Q_3 ; during a left shift, 0 enters Q_0 . Initially, $Q_3 Q_2 Q_1 Q_0 = 1011$. The control inputs on four successive positive clock edges are

01, 10, 00, 01.

The final contents of the register are

(A) 0010 (B) 0100 (C) 0101 (D) 1010

Q3.

[1 Mark]

A 4-bit register consists of positive-edge-triggered D flip-flops. The next-state equations are

$$Q_3^+ = Q_2, \quad Q_2^+ = Q_1, \quad Q_1^+ = Q_0, \quad Q_0^+ = Q_3 \oplus Q_2.$$

Starting from $Q_3 Q_2 Q_1 Q_0 = 0001$, the number of distinct states visited, **including the initial state**, before the register first returns to 0001 is

(A) 4 (B) 7 (C) 8 (D) 15

Q4.

[1 Mark]

During the transition of a 4-bit asynchronous up counter, in which each higher-order flip-flop is triggered by the output transition of the preceding flip-flop, from the stable state $0111 \rightarrow 1000$, the following states are momentarily observed:

$$0111 \rightarrow 0110 \rightarrow 0100 \rightarrow 0000 \rightarrow 1000.$$

Assume that all flip-flops have equal, nonzero propagation delay. Which one of the following best explains the appearance of these intermediate states?

- (A) The flip-flops receive the same clock edge, but differences in their setup times cause them to change one after another.
- (B) Each flip-flop is clocked by the output of the preceding flip-flop, so the state change propagates sequentially due to the propagation delay of each flip-flop.

- (C) The counter temporarily enters invalid states because more than one flip-flop is not allowed to change during the same count transition.
- (D) The intermediate states are caused by a race condition, and their order cannot be predicted even when all flip-flops have equal propagation delay.

Q5.

[1 Mark]

Consider a 4-bit carry lookahead adder, where

$$G_i = A_i B_i, \quad P_i = A_i \oplus B_i, \quad C_{i+1} = G_i + P_i C_i.$$

For

$$(G_3 G_2 G_1 G_0) = 0100, \quad (P_3 P_2 P_1 P_0) = 1011, \quad C_0 = 1,$$

which one of the following gives the correct values of (C_1, C_2, C_3, C_4) ?

- (A) (1, 1, 1, 1) (B) (1, 1, 1, 0) (C) (1, 1, 0, 0) (D) (0, 1, 1, 0)

Q6.

[3 Marks]

A combinational circuit implements

$$F(A, B, C) = \overline{A}B + AC.$$

Initially, $(A, B, C) = (1, 1, 1)$, and only A changes from 1 to 0. Ideally, F should remain equal to 1 throughout the transition. However, the physical circuit may momentarily produce $F = 0$.

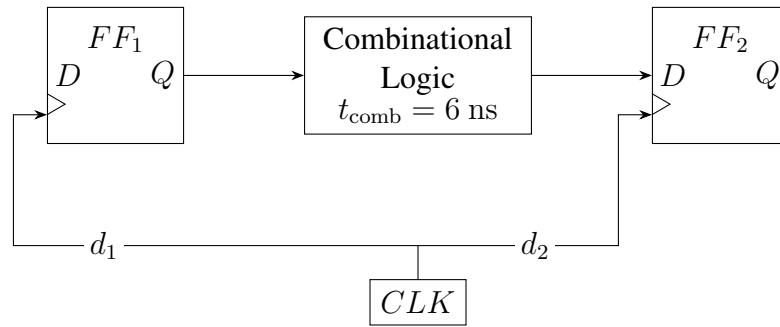
- (a) Explain this momentary change in F using a qualitative timing waveform.
- (b) Without changing the Boolean function, modify the expression so that this glitch is eliminated. Briefly explain why the modification works.

Answer:

Q7.

[2 Marks]

Consider the synchronous circuit shown below. The same clock source drives two positive-edge-triggered D flip-flops through different clock-distribution paths.



For both flip-flops: $t_p = 2 \text{ ns}$, $t_{\text{setup}} = 1 \text{ ns}$, $t_{\text{hold}} = 0.5 \text{ ns}$

For the two clock-distribution designs,

	d_1	d_2
Design I	3 ns	5 ns
Design II	5 ns	3 ns

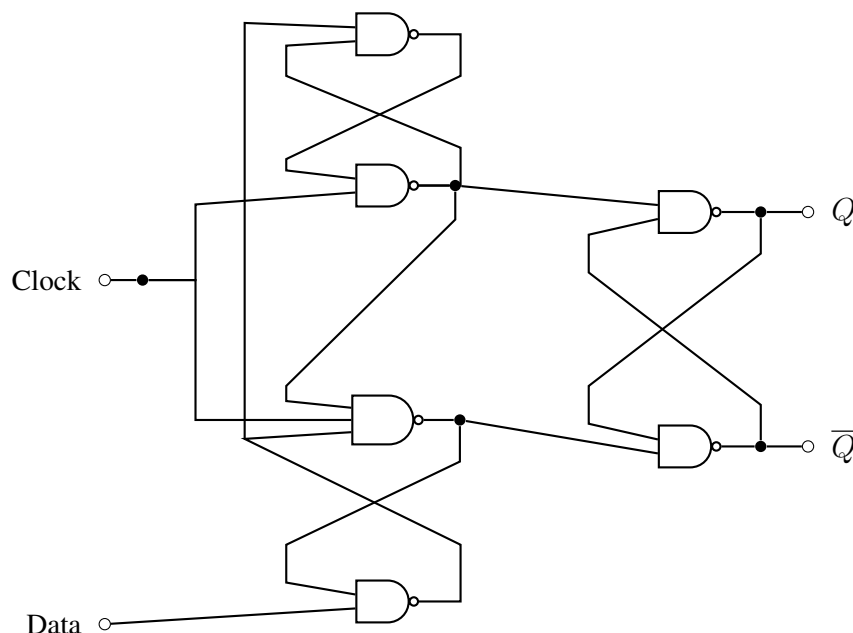
Assume that the hold-time constraint is satisfied in both designs. Determine the minimum safe clock period for Design I and Design II.

Answer:

Q8.

[5 Marks]

The following gate-level circuit implements a **positive-edge-triggered D flip-flop**.



Study the circuit carefully and answer the following:

- (a) Suppose $D = 1$ just before the positive edge of CLK . Explain how the circuit stores $Q = 1$ at the positive clock edge.
- (b) After the positive edge, CLK remains HIGH and D changes from 1 to 0. Explain why Q does not change during the same HIGH clock level.
- (c) A master–slave flip-flop also prevents the input from continuously propagating to the output. However, the master latch remains transparent during one clock level and may capture a temporary change in D during that interval. Explain how the edge-triggered implementation shown above avoids this *0/1-catching* problem.

Answer: